VIT Vidyalankar Institute of Technology Accredited A+ by NAAC	Department of Computer Engineering A.Y-2026-27

Semester	B.E. Semester VII – CMPN
Subject	DevOps
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Assisting Teachers	Prof. Snehal Andhare
Laboratory	

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Grade and Subject Teacher's Signature		

Experiment Number	8
Experiment Title	Luggage Complaint Portal: Pipeline as Code and Server Deployment
Objectives (Skill Set / Knowledge Tested / Imparted)	To create a Jenkinsfile with checkout, build, package and deploy stages, deploy the build to Tomcat and parameterise an environment setting.
Program	

1. Tomcat installed as the deployment target

Apache Tomcat 10.1 was installed in /opt/tomcat, set to port 8082 (Jenkins uses 8080) and run as a systemd service. The jenkins user was added to the tomcat group so that it can copy the WAR into webapps.

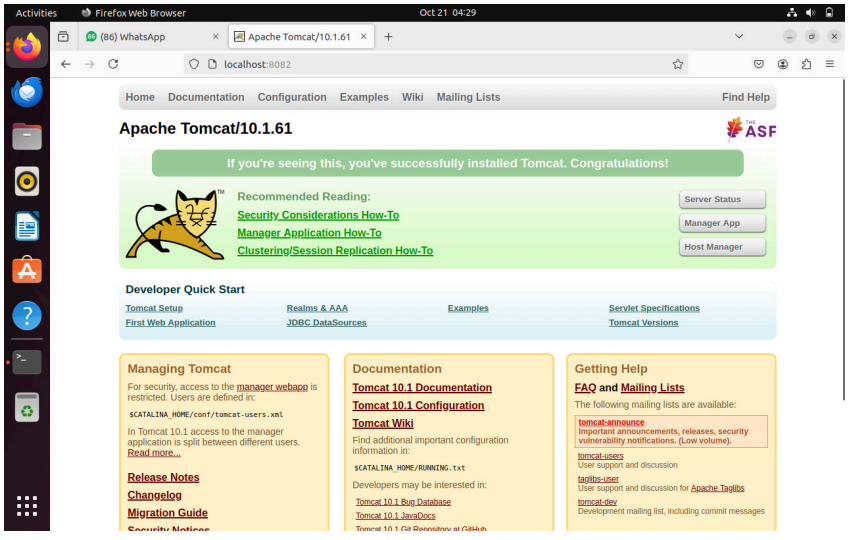


Figure: Tomcat 10.1 welcome page on port 8082

2. Jenkinsfile (pipeline as code)

```
pipeline {
    agent any
    parameters {
        choice(name: 'DEPLOY_ENV', choices: ['dev', 'staging'],
               description: 'Environment label baked into the app')
    }
    environment {
        JAVA_HOME = '/usr/lib/jvm/java-17-openjdk-amd64'
```

```
        WEBAPPS    = '/opt/tomcat/webapps'
        APP_URL    = 'http://localhost:8082/luggage-portal'
        APP_ENV    = "${(params.DEPLOY_ENV ?: 'dev').toUpperCase()}"
    }
    stages {
        stage('Checkout') { steps { checkout scm } }
        stage('Build')    { steps { sh 'mvn -B clean compile' } }
        stage('Package')  { steps { sh 'mvn -B package -DskipTests
-Dapp.env=$APP_ENV -Dapp.build=$BUILD_NUMBER' } }
        stage('Deploy')   { steps { sh 'cp target/luggage-portal.war
$WEBAPPS/luggage-portal.war' } }
    }
}
```

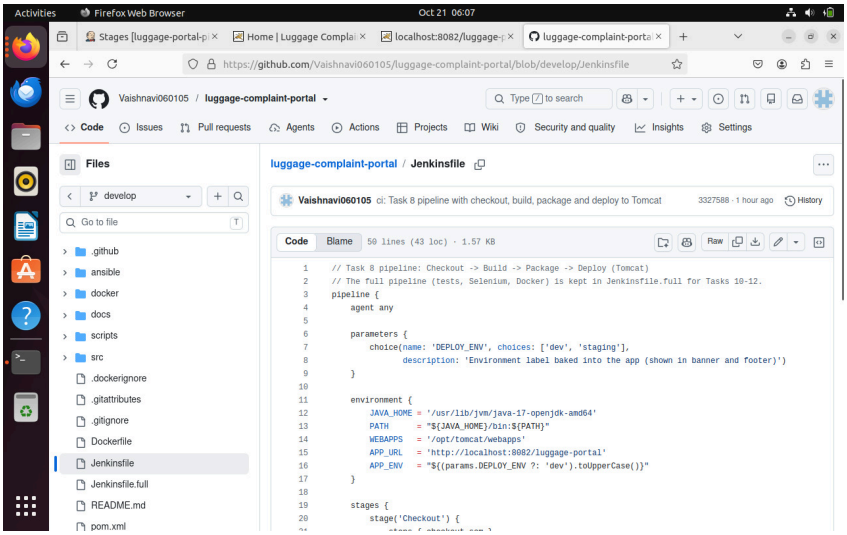


Figure: Jenkinsfile on GitHub (develop branch)

3. Pipeline run

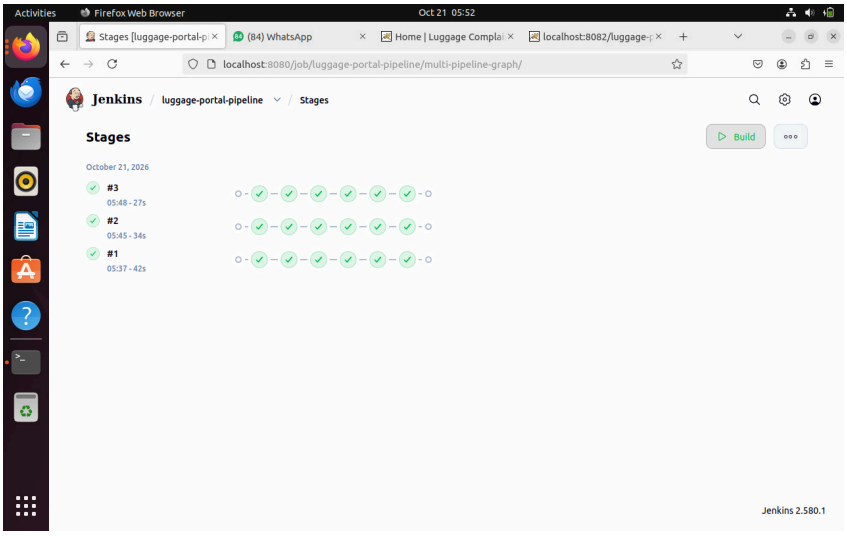


Figure: Stages view of builds #1 to #3, all stages green

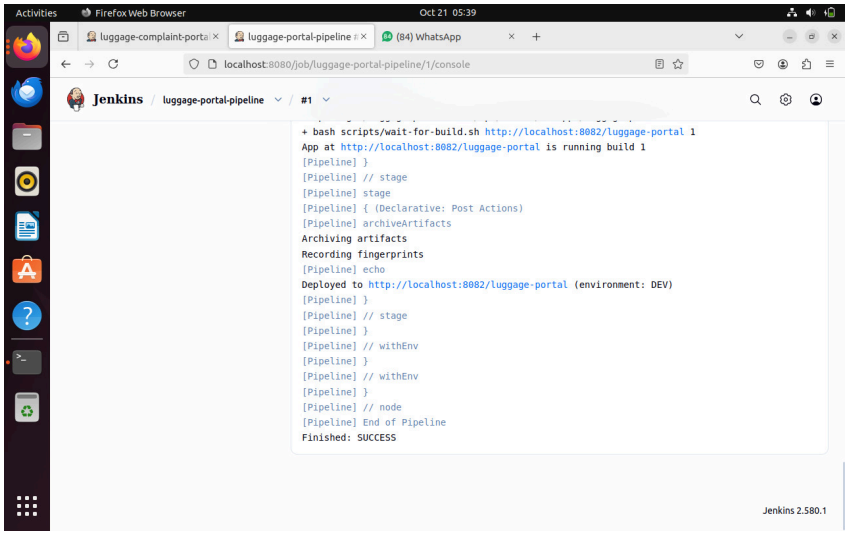


Figure: Console output: deployed to http://localhost:8082/luggage-portal (environment: DEV)

4. Parameterised environment setting

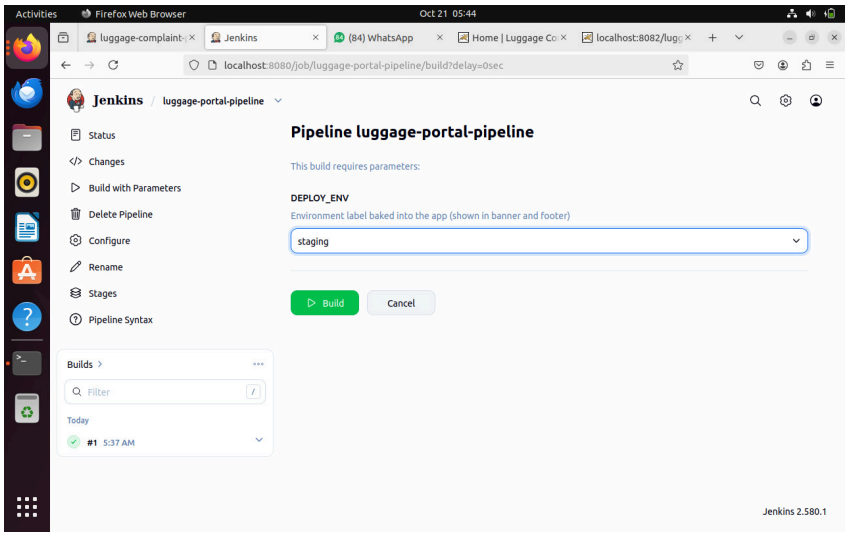


Figure: Build with Parameters: DEPLOY_ENV = staging

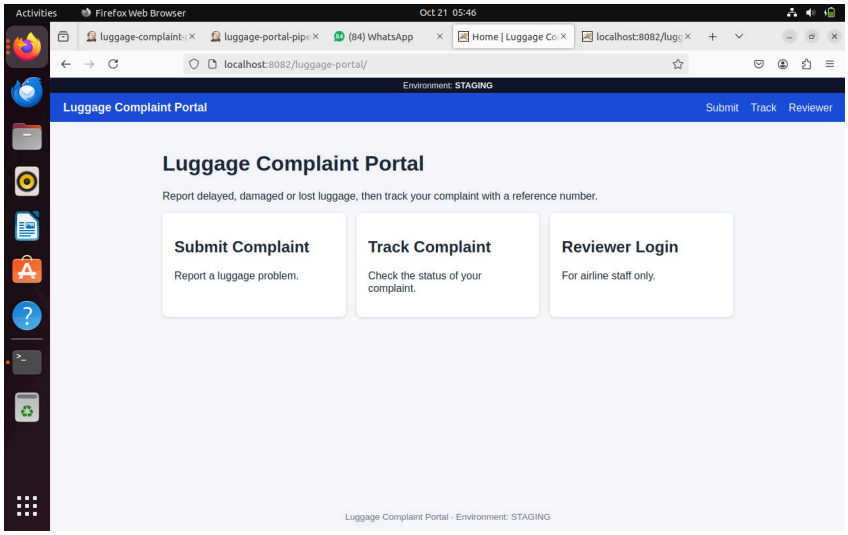


Figure: Deployed application showing Environment: STAGING

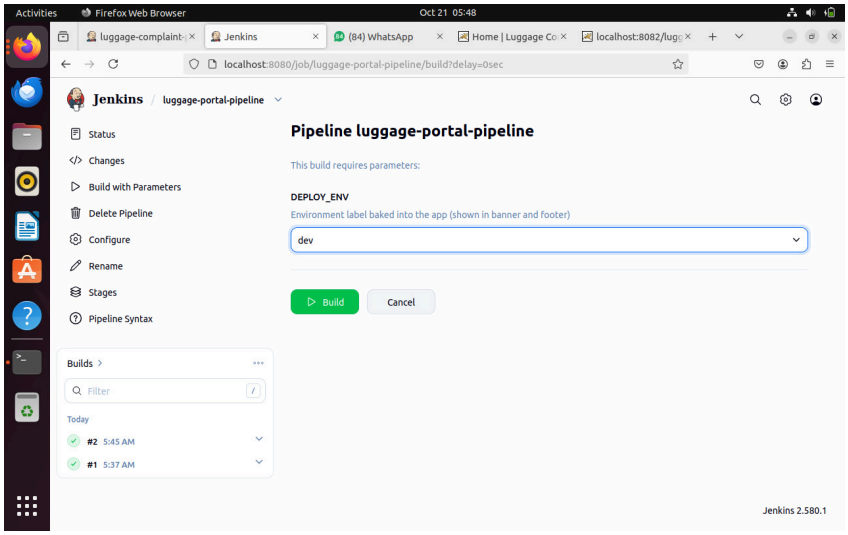
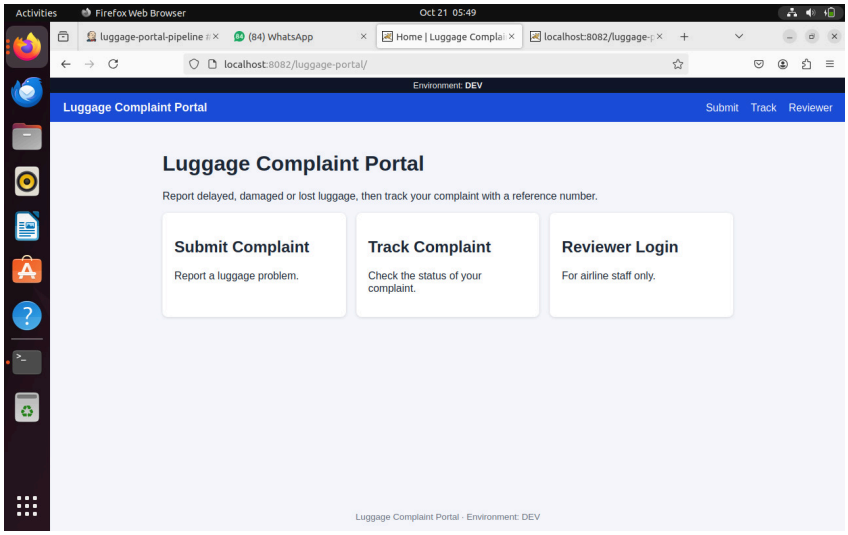


Figure: Build with Parameters: DEPLOY_ENV = dev

	 *Figure: Deployed application showing Environment: DEV*
Conclusion	A Jenkinsfile with checkout, build, package and deploy stages deployed the application to Tomcat, and the DEPLOY_ENV parameter changed the environment shown in the application.